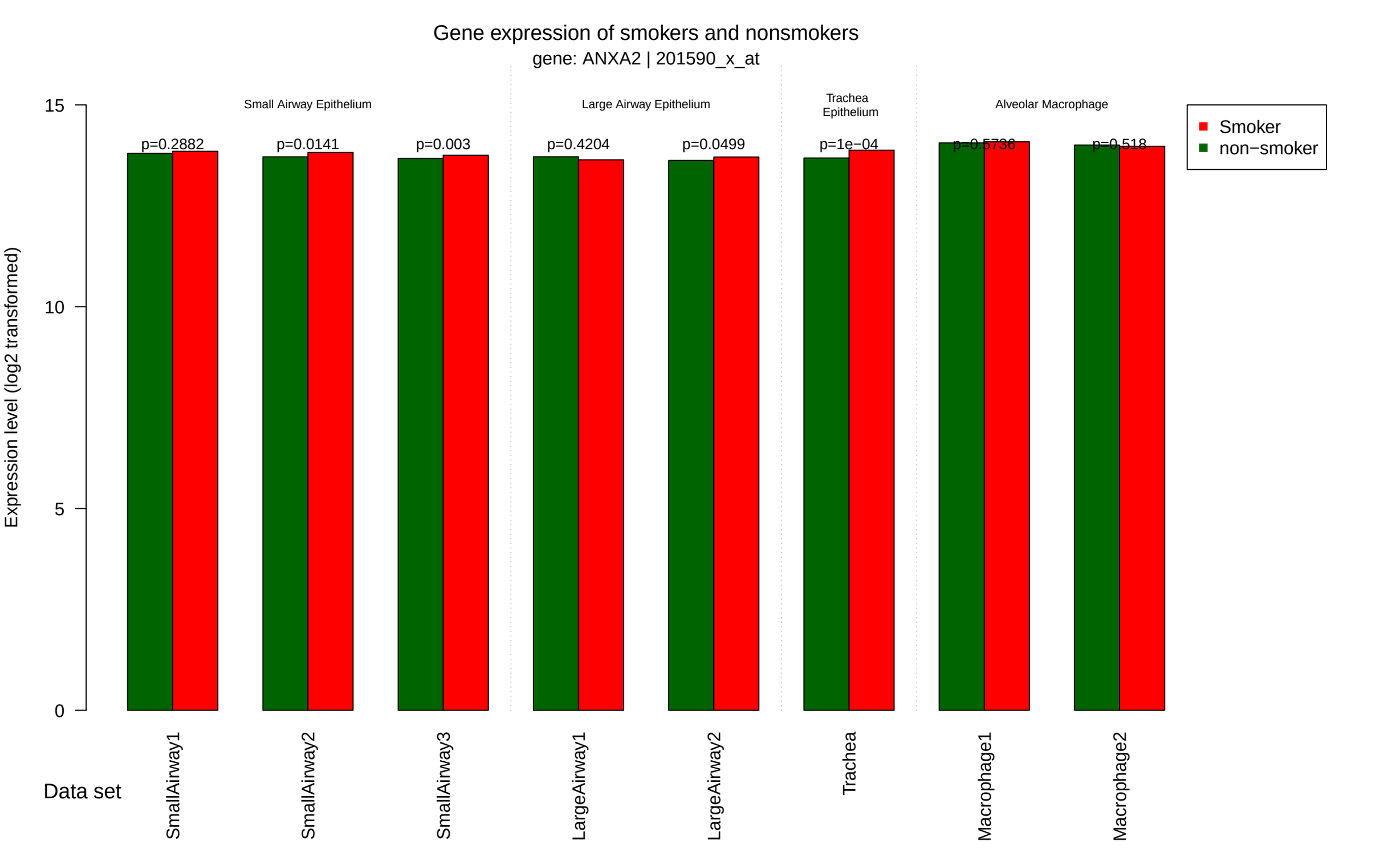
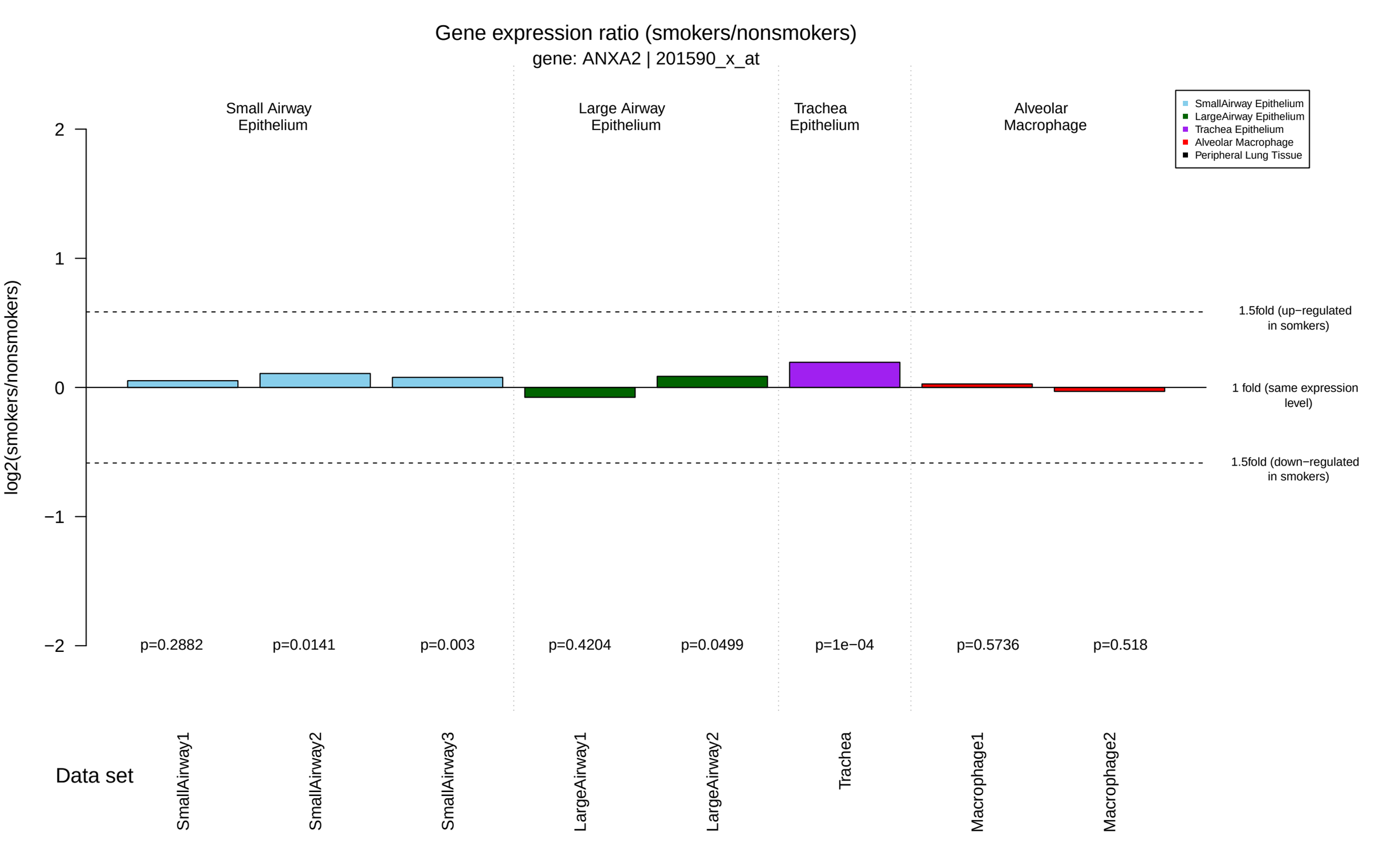
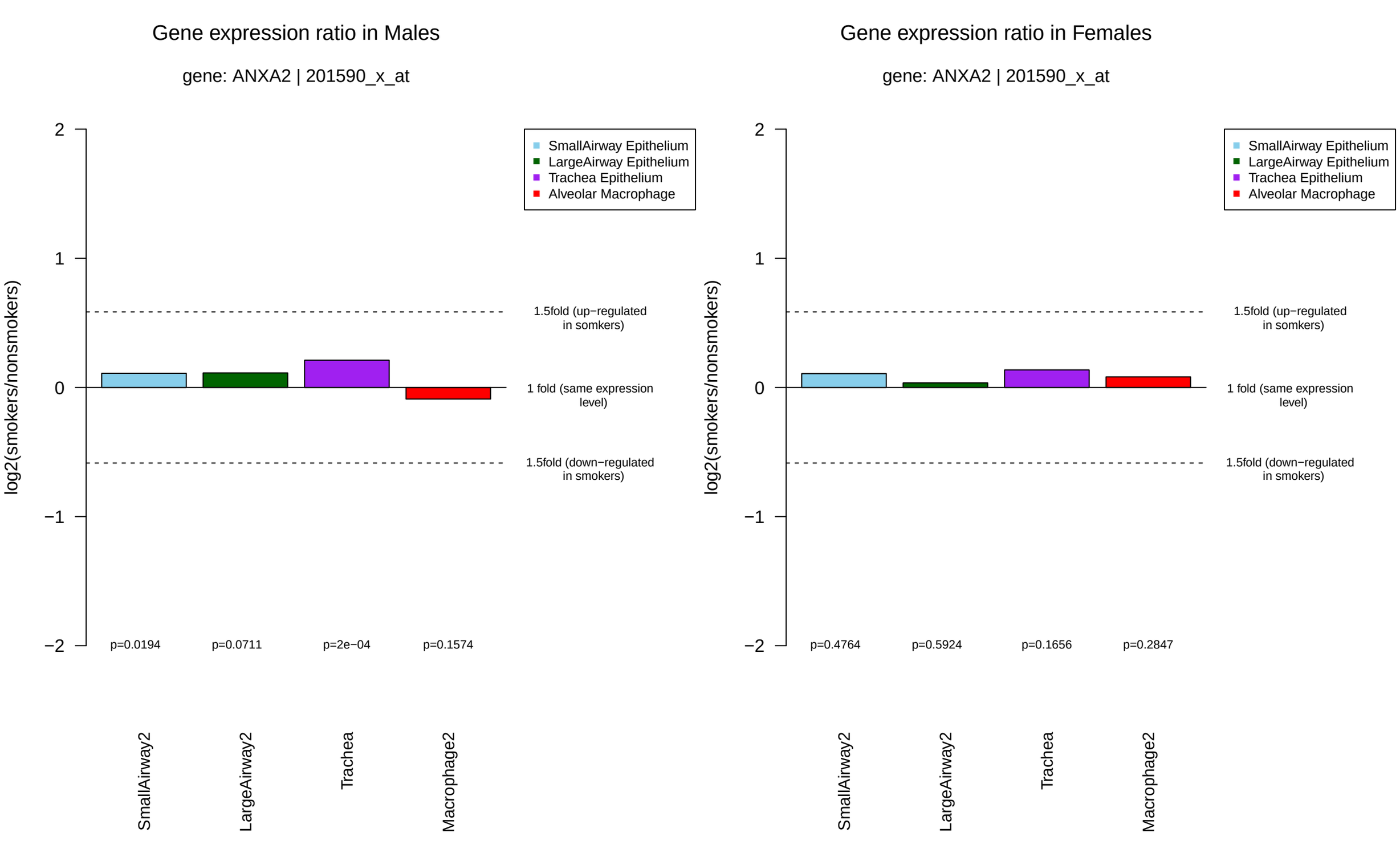
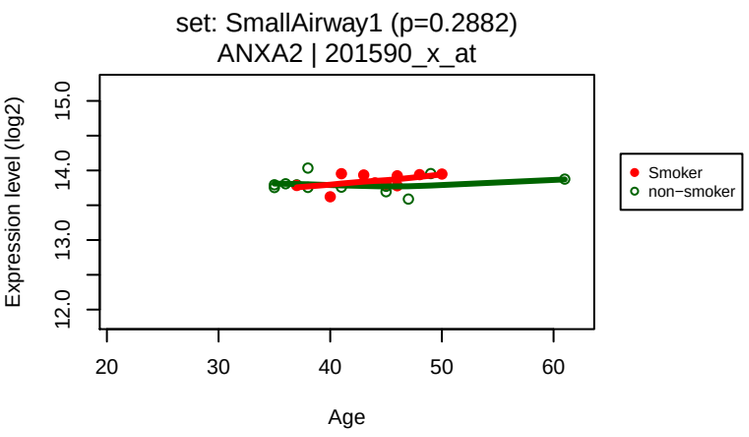
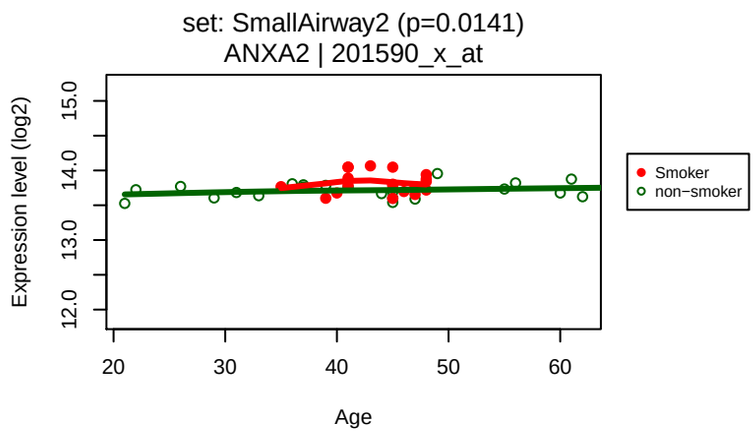
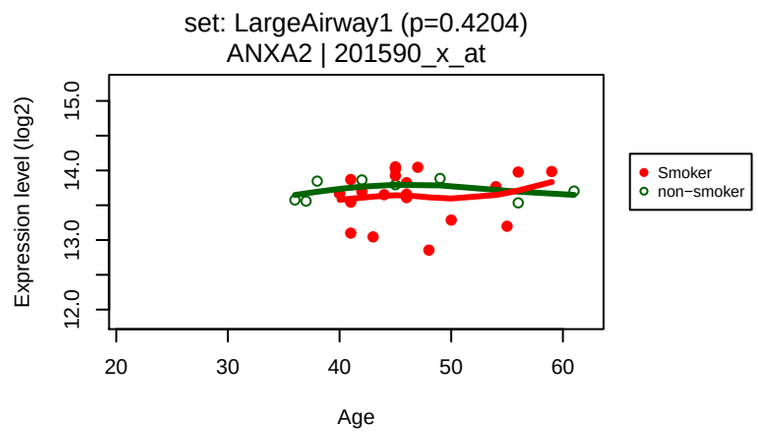

Gene expression of smokers and nonsmokers
gene: ANXA2 | 201590_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2882
p=0.0141
p=0.003
p=0.4204
p=0.0499
p=1e−04
p=0.5736
p=0.518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

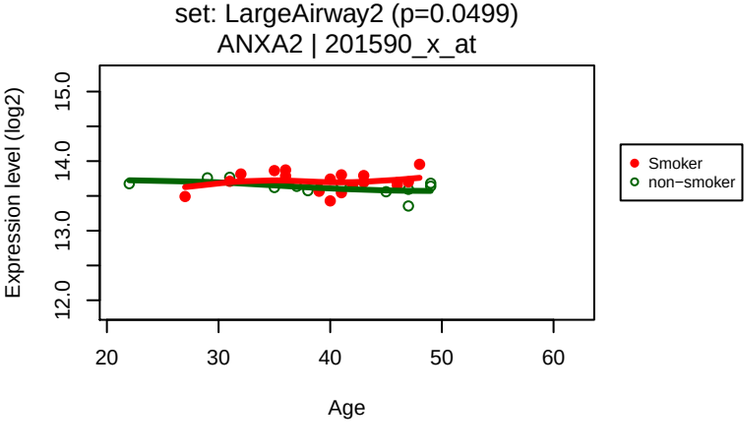

Gene expression ratio (smokers/nonsmokers)
gene: ANXA2 | 201590_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2882
p=0.0141
p=0.003
p=0.4204
p=0.0499
p=1e−04
p=0.5736
p=0.518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

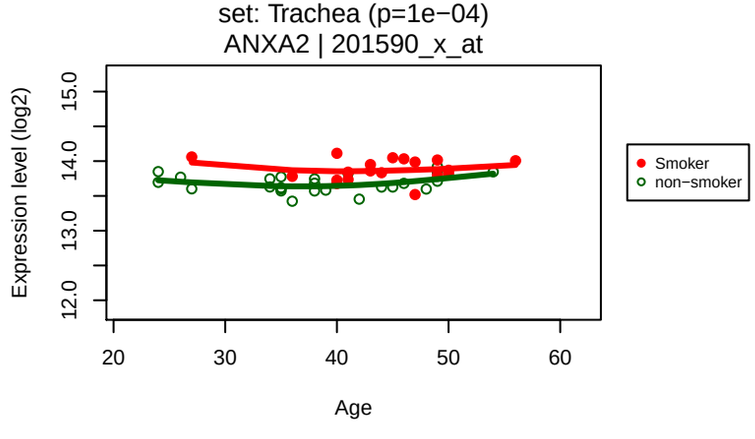

Gene expression ratio in Males
gene: ANXA2 | 201590_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0194
p=0.0711
p=2e−04
p=0.1574
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANXA2 | 201590_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4764
p=0.5924
p=0.1656
p=0.2847
SmallAirway2
LargeAirway2
Trachea
Macrophage2

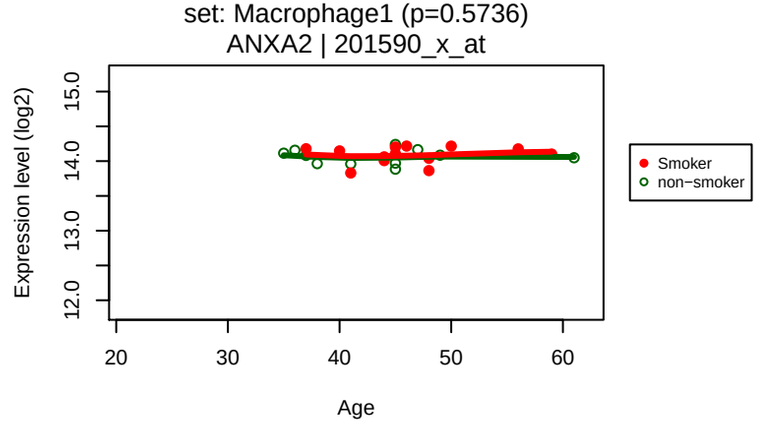

set: SmallAirway1 (p=0.2882)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

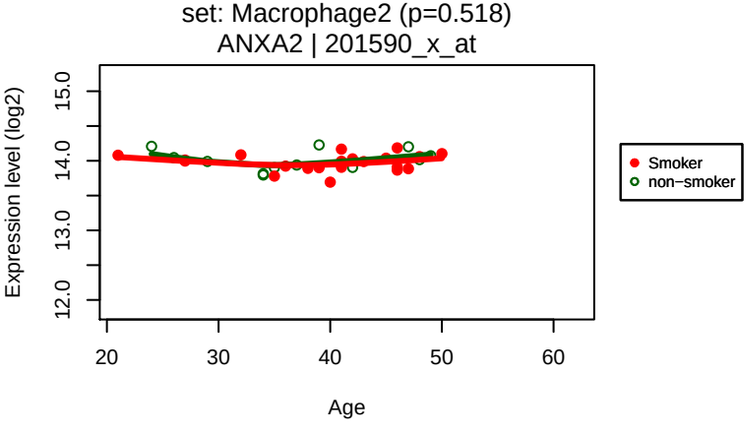

set: SmallAirway2 (p=0.0141)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

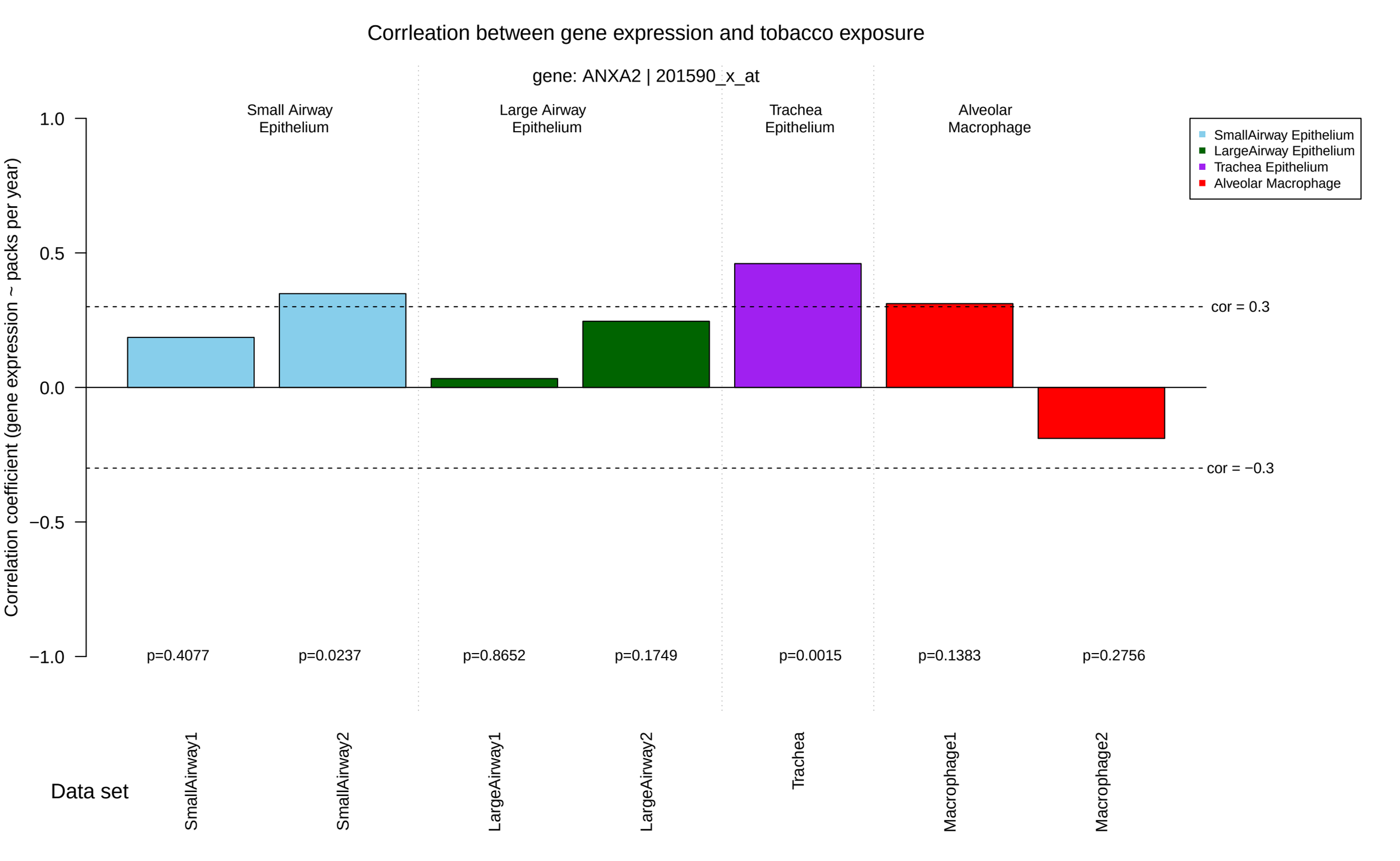

set: LargeAirway1 (p=0.4204)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0499)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5736)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.518)
ANXA2 | 201590_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANXA2 | 201590_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4077
p=0.0237
p=0.8652
p=0.1749
p=0.0015
p=0.1383
p=0.2756
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2